Notes: Liu, Lu, and Yuan (JDE, 2025)

The Quiet Revolution: Send-down Movement and Female Empowerment in China

Contents

1 Big picture

- **Question.** What promotes female empowerment and gender equality in traditional societies? More specifically, can temporary internal population mobility transmit gender-equal ideas and change the long-run choices of rural women?
- **Setting.** The paper studies China's Send-down Movement, a large government-mandated urban-to-rural relocation program from the late 1960s through the 1970s. More than 16 million urban youths, called sent-down youth (SDYs), were sent to rural counties.
- **Core historical fact.** Urban women experienced earlier progress in education and employment under Mao-era gender-equality ideology, while rural women remained more constrained by traditional patriarchal norms. The movement temporarily brought urban youth, education, and social values into rural villages.
- **Main conceptual idea.** SDYs affected rural girls through two channels. First, many SDYs became primary school teachers and improved rural human capital formation. Second, SDYs brought urban, more gender-equal beliefs into daily school, work, and social interactions.
- **Empirical challenge.** Counties that received more SDYs may differ from counties that received fewer SDYs. The paper therefore does not rely only on cross-county comparisons; it uses within-county birth-cohort variation in whether rural children were of primary-school age during the movement.
- **Main empirical design.** The baseline design is a cohort difference-in-differences model. Treatment cohorts are rural individuals born in 1956–1969, who were at primary-school age during the movement. Control cohorts are those born in 1946–1955, who had already passed primary-school age before SDYs arrived. Treatment intensity is the county-level population share of SDYs.
- **Education and labor result.** A two-percentage-point increase in the SDY population share increased female middle school completion by about 0.868 percentage points and high school completion by about 1.29 percentage points. The corresponding effects for men were smaller, so gender gaps narrowed.
- **Employment and financial autonomy result.** SDY exposure raised rural female formal labor participation, weekly hours, and reliance on one's own work as the main source of income, while reducing reliance on family support. Again, the gains were larger for women than for men.
- **Marriage, fertility, and politics result.** Greater SDY exposure delayed rural women's first marriage, increased divorce and never-married probabilities, reduced fertility, and increased female political participation, including Communist Party membership and supportive views of civic participation.
- **Ideology result.** Exposed rural women reported stronger preferences for formal employment, financial independence, independent problem-solving, self-confidence, willingness to take risk, equal spousal relations, and lower desired fertility.
- **Mechanism result.** Longer in-school exposure to SDYs generated larger gains in education and labor-market outcomes, consistent with human capital accumulation. But gender-equal attitudes changed even with shorter exposure, suggesting that ideology and aspiration transmission can occur relatively quickly.

- **Economic takeaway.** The paper argues that population mobility can do more than move labor. It can also move social values. Temporary contact between urban migrants and rural communities helped transmit gender-equal norms and generated persistent gains in rural female empowerment.

2 Data and measurement (key objects)

2.1 Why this setting is useful

- The Send-down Movement was a massive, temporary, state-mandated migration shock. Rural counties received different intensities of SDY inflows, and the timing of exposure varied mechanically by birth cohort.
- SDYs were generally young urban people with higher education than rural peers. Many became rural primary-school teachers, while others worked with rural residents in agricultural and social settings.
- Rural China before the movement had strong gender inequality, especially in education, formal employment, and family decision-making. This makes the setting useful for studying whether outside contact can weaken entrenched gender norms.
- The movement was temporary, ending after Mao's death, but the paper studies outcomes decades later. The setting therefore speaks to long-run effects of early-life social interactions.

Interpretation. The paper uses the movement as a quasi-experimental social-contact shock. The key hypothesis is not simply that rural schools received more teachers, but that rural girls encountered urban youth who embodied different beliefs about women's social roles.

2.2 Institutional background

- **Pre-communist and early communist China.** Rural women historically faced male-biased norms, limited education, arranged marriage, economic dependence, and restricted autonomy.
- **Urban-rural divergence.** Communist-era pro-women policies and slogans such as "women hold up half the sky" advanced women's status, but many gains were concentrated in urban areas. Rural areas retained stronger traditional norms.
- **Send-down Movement.** Starting in 1968, the government sent urban youth to rural areas to address urban unemployment and political unrest. SDYs were assigned across rural counties and often returned after the policy was reversed around 1978–1979.
- **Rural interaction channels.** In schools, SDYs could act as better-educated teachers. Outside schools, female SDYs and urban youth could serve as role models and transmit gender-equal views through work, host-family, and village interactions.

2.3 Main treatment variable

The main county-level treatment intensity is:

$$SDY_c = \frac{\text{number of SDYs received by county } c \text{ during 1968–1977}}{\text{county } c \text{ population in 1964}}.$$

- The numerator is compiled from local county gazetteers and related historical records.
- The denominator is pre-movement local population, so the variable measures the intensity of SDY exposure relative to the county's initial size.
- The average SDY population share is about 2% of the local population, but SDYs represented a much larger share of the local young adult population.

Interpretation. SDY_c is the cross-county intensity of contact with urban youth. The paper interacts this intensity with cohort exposure to isolate whether cohorts young enough to interact with SDYs in primary school changed more in high-SDY counties.

2.4 Treatment and control cohorts

- **Treatment group:** individuals born in 1956–1969. They were of primary-school age during some part of the movement and had frequent potential contact with SDYs in school.
- **Control group:** individuals born in 1946–1955. They had generally passed primary-school age before the movement and therefore had less school-based SDY exposure.
- The paper also uses broader birth cohorts in event-study plots and robustness checks.

Interpretation. The design compares younger and older cohorts within the same county. A high-SDY county should matter more for cohorts whose primary schooling overlapped with SDY arrival.

2.5 Main data sources

- **County gazetteers and historical records:** counts of SDYs received by counties during the movement.
- **Population Census 2000:** primary source for education, labor participation in 2000, marriage outcomes, and rural demographic characteristics.
- **Population Census 2010:** labor supply, weekly working hours, main source of income, financial independence, and fertility history.
- **CHIP surveys:** detailed employment outcomes and attitudes toward work, independence, confidence, risk, and life aspirations.
- **CFPS surveys:** Party membership, social values, mental health, happiness, and life satisfaction.
- **CSS surveys:** political and civic participation attitudes and beliefs about government responsibility.

- **County controls:** pre-movement geography, agricultural output, historical gender norms, Cultural Revolution intensity, Confucian academy density, wartime base indicators, migration measures, and other local characteristics.

2.6 Outcome variables

- **Education:** years of schooling, primary school completion, middle school completion, and high school completion.
- **Labor market:** current work status, weekly working hours, formal non-agricultural work, paid employment, public/private sector work, and labor participation in the Census.
- **Financial independence:** whether the individual's main income source is own work versus family support.
- **Marriage and fertility:** age at first marriage, ever divorced, never married, and number of children.
- **Political empowerment:** Communist Party membership and survey responses about civic participation, petitioning, strikes/boycotts, and government responsibility.
- **Ideologies and self-perceptions:** value of formal employment, financial independence, overcoming challenges independently, self-confidence, willingness to take risk, equal spousal relations, desired number of children, and views on traditional norms.
- **Later-life welfare:** mental health using the K6 scale, depression indicators, happiness, life satisfaction, and positive emotions.

2.7 Descriptive facts

- In the main Census sample, rural females born in 1946–1969 average about 6.85 years of schooling. About 88.1% completed primary school, 41.0% completed middle school, and 9.3% completed high school.
- Rural males in the same cohorts have higher average schooling, with about 96.7% completing primary school, 62.6% completing middle school, and 16.7% completing high school.
- Female labor participation is high, but financial dependence remains gendered. In 2010, about 78.0% of rural females and 92.5% of rural males relied mainly on their own work as income.
- The control cohorts already had high primary completion, so the paper focuses more on middle and high school completion, where there was room for gender-gap narrowing.

Interpretation. The data show a large baseline gender gap in schooling and labor-market autonomy. This makes middle school completion, high school completion, and own-work income especially informative measures of empowerment.

3 Models and empirical specifications (all equations + interpretation)

3.1 Baseline cohort difference-in-differences model

The baseline individual-level specification is:

$$Y_{igc} = \alpha_0 + \alpha_1 SDY_c \times Treat_g + \alpha_2 X_i + \lambda_c + \mu_{pg} + \varepsilon_{igc}.$$

- Y_{igc} is an outcome for individual i born in cohort g in county c.
- SDY_c is the county's SDY population share.
- $Treat_g$ equals one for cohorts born in 1956–1969 and zero for cohorts born in 1946–1955.
- X_i includes individual controls, especially Han ethnicity.
- λ_c are county fixed effects.
- μ_{pg} are province-by-cohort fixed effects.
- Standard errors are clustered at the county level.

Interpretation. The coefficient α_1 asks whether treatment cohorts experienced larger changes in counties with more SDY inflow, relative to older cohorts in the same county and relative to cohort trends within the same province.

3.2 Triple-interaction specification

To test whether SDY exposure affected women more than men, the paper estimates:

$$\begin{aligned} Y_{igc} = \beta_0 &+ \beta_1 Female_i + \beta_2 SDY_c \times Treat_g \\ &+ \beta_3 SDY_c \times Treat_g \times Female_i + \beta_4 X_i + \lambda_c + \mu_{pg} \\ &+ \beta_5 Female_i \times X_i + Female_i \times \lambda_c + Female_i \times \mu_{pg} + \varepsilon_{igc}. \end{aligned}$$

Interpretation. The coefficient β_3 is the female-minus-male difference in the treatment effect. A positive β_3 means that SDYs improved the outcome more for rural women than for rural men, after allowing gender-specific controls and fixed effects.

3.3 Gender-gap specification

The paper also estimates county-cohort regressions directly on gender gaps:

$$GenderGap_{gc} = \gamma_0 + \gamma_1 SDY_c \times Treat_g + \gamma_2 X_{gc} + \lambda_c + \mu_{pg} + \varepsilon_{gc}.$$

- $GenderGap_{gc}$ is measured either as the female-to-male ratio or as the female-minus-male difference for a given outcome.
- X_{gc} includes county-cohort gender and ethnicity composition controls.

- Regressions are weighted by county-cohort population size.

Interpretation. This specification directly tests whether high-SDY counties experienced larger improvements in gender equality for exposed cohorts.

3.4 Event-study specification

To test pre-trends, the paper estimates:

$$Y_{igc} = \alpha_0 + \sum_{\gamma=1946,\, \gamma\neq 1954/1955}^{1979} \alpha_\gamma SDY_c \cdot \mathbf{1}\{g=\gamma\} + \alpha_2 X_i + \lambda_c + \mu_{pg} + \varepsilon_{igc}.$$

Interpretation. The event-study coefficients show whether high-SDY and low-SDY counties were already trending differently for older cohorts before the movement. Flat pre-treatment coefficients support the cohort-DID design.

3.5 Mechanism tests

The paper separates human capital accumulation from ideology transmission using two complementary approaches.

- **In-school intensity.** Treatment cohorts are split by the number of primary-school years spent with SDYs. High-intensity cohorts had 5–6 years of overlap, while low-intensity cohorts had 1–4 years.
- **Prediction.** If the outcome is knowledge-based, longer exposure should matter more. If the outcome is ideology transmission, even short exposure may matter and additional years may have smaller marginal effects.
- **Out-of-school interactions.** The paper studies rural women who had already passed primary-school age but were close in age to SDYs and therefore likely interacted with them in work and social settings.
- **Tea cultivation test.** Because tea cultivation uses more female labor, female SDYs and rural women should interact more in tea-growing areas. Orchard cultivation, which uses more male labor, serves as a placebo comparison.

Interpretation. The mechanism section asks not only whether SDYs changed outcomes, but whether the changes came from schooling, role-model effects, social contact, or a mix of these channels.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the baseline effect

- Identification comes from comparing exposed and older cohorts within the same county, while allowing province-specific cohort trends.
- County fixed effects absorb time-invariant county differences, such as location, historical gender norms, and average development.
- Province-by-cohort fixed effects absorb broad cohort-specific shocks within each province.
- The treatment is stronger in counties with higher SDY population shares, but only for cohorts young enough to interact with SDYs in primary school.

Core identifying assumption. In the absence of the Send-down Movement, high-SDY and low-SDY counties would have followed similar cohort trends in female empowerment outcomes, conditional on fixed effects and controls.

4.2 Why the assignment is plausibly useful

- The paper emphasizes that the identification strategy does not require SDY inflows to be perfectly randomly assigned across counties.
- What matters is that SDY intensity is not correlated with differential cohort trends in female outcomes before the movement.
- The authors check that SDY population share is not systematically correlated with pre-movement county characteristics such as agricultural output, geography, extreme weather, female education, female labor participation, marriage, fertility, and bargaining proxies.

4.3 Parallel trends and event-study evidence

- Event-study plots show small and statistically insignificant coefficients for pre-treatment cohorts.
- Effects become positive mainly for cohorts whose primary-school years overlap with the movement.
- The pattern is stronger for rural women than for rural men and is visible both in individual outcomes and in county-cohort gender gaps.

Interpretation. The event-study evidence supports the idea that the estimated effects are not driven by pre-existing cohort trends in high-SDY counties.

4.4 Alternative explanations addressed

- **Harsh destination selection.** SDYs might have been sent to poorer or harsher agricultural areas. The paper shows little systematic correlation between SDY share and pre-movement agricultural or geographic conditions.
- **Other historical shocks.** Results are robust to accounting for the Great Famine, Cultural Revolution intensity, Family Planning policies, market-oriented reforms, the Third Front construction campaign, and other local historical factors.

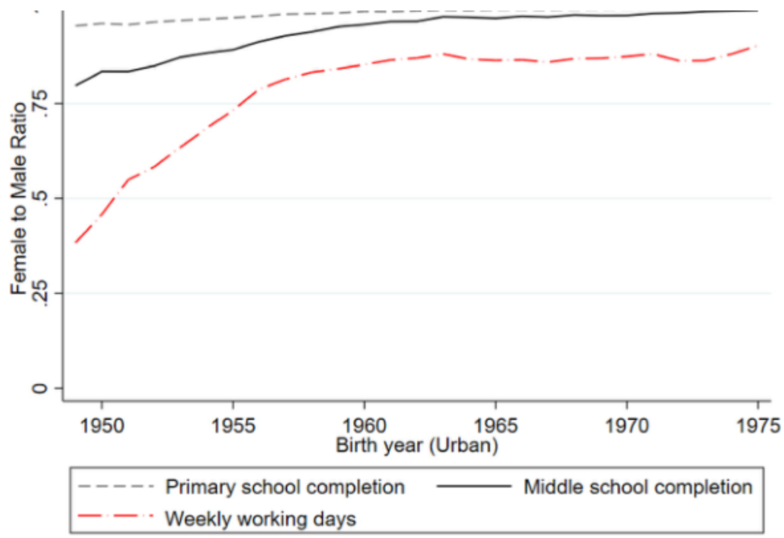
- **Later migration.** SDY share is not systematically related to county in-migration or out-migration in the mid-1990s, and results are robust to excluding high-migration counties or controlling for migration-related cohort trends.
- **Urban spillovers.** The paper finds no comparable effect on urban gender equality, which supports a direct rural exposure interpretation.
- **Gender composition of SDYs.** The gender balance of SDYs appears roughly even and not strongly correlated with recipient-county characteristics.
- **Permutation tests.** Randomly reassigning SDY intensity across counties generates insignificant placebo coefficients.
- **Alternative treatment cohorts.** Results are robust to alternative definitions of exposure based on different school starting and ending ages.

4.5 What the paper can and cannot fully separate

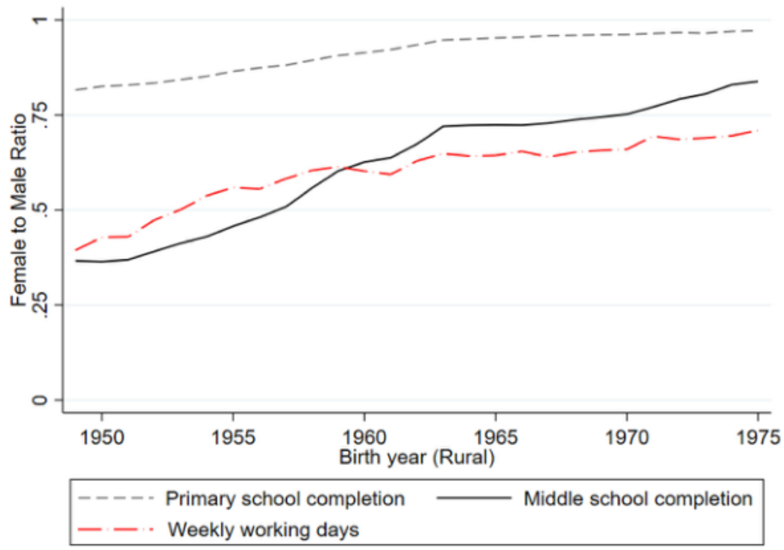
- The paper provides strong evidence that SDYs affected rural female outcomes, and the mechanism evidence points to both schooling and ideology transmission.
- The design cannot literally randomize which rural girl interacted with which SDY teacher or coworker, so individual-level mechanisms remain partly inferred from timing and heterogeneity tests.
- The paper is especially persuasive on the combined effect of SDY exposure, while the exact decomposition between human capital, role-model effects, and social-contact effects is more suggestive.

5 Tables and figures

5.1 Figure 1: Gender gap in urban versus rural China



(A) **Gender Gap in Urban Population**



(B) **Gender Gap in Rural Population**

Figure 1: Gender gap in education and labor market outcomes in urban and rural China.

Read. The figure plots female-to-male ratios in primary school completion, middle school completion, and weekly working days for urban and rural cohorts. Urban women approach parity earlier. Rural women improve sharply for cohorts born in the late 1950s and 1960s, but gender gaps remain larger than in urban areas.

Economic message. The figure motivates the paper. Rural China experienced a sharp improvement in female outcomes during exactly the cohorts exposed to the Send-down Movement, creating a puzzle: what drove rapid female empowerment in a traditional rural setting?

5.2 Figure 2: Geographic distribution of SDY exposure

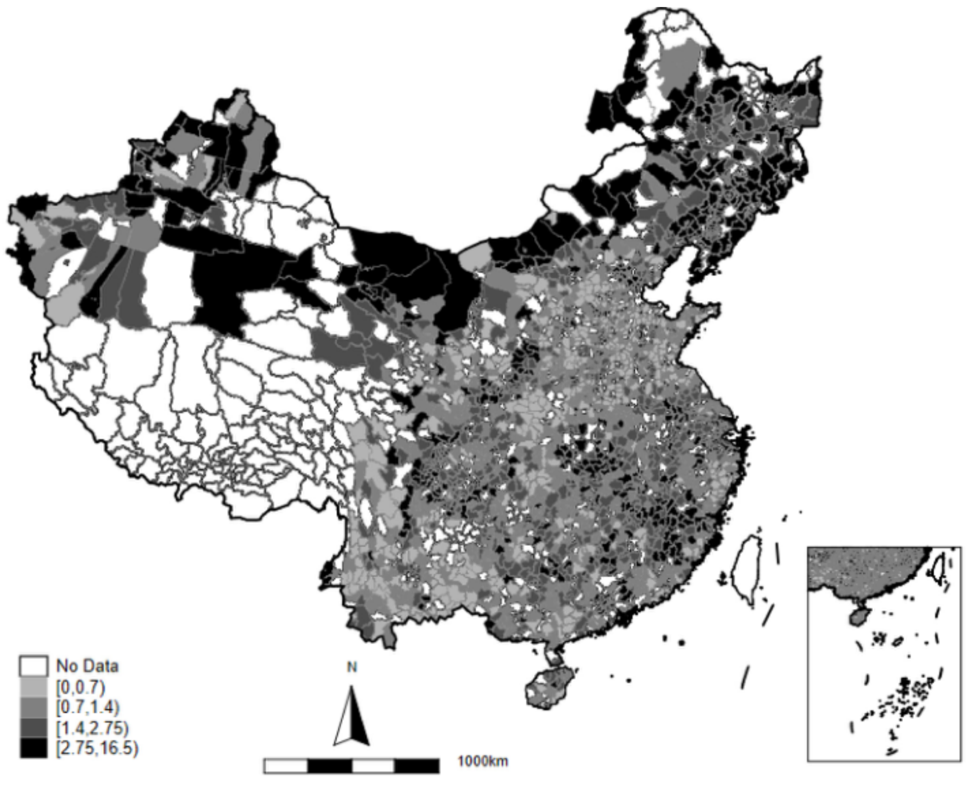


Figure 2: County-level population share of SDY inflows during 1968–1977.

Read. The map shows substantial variation in SDY population shares across rural counties, without an obvious single regional cluster driving the variation.

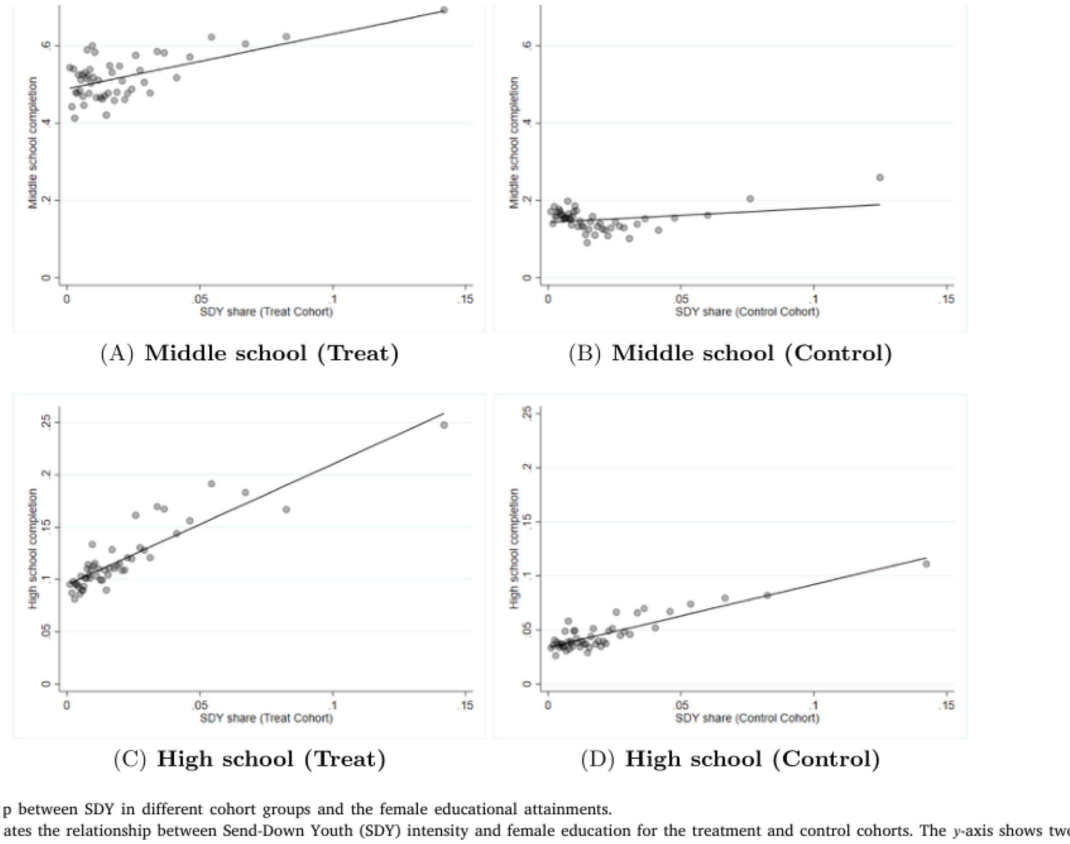
Economic message. Cross-county treatment intensity is essential for the design. The map shows that different rural counties had different degrees of contact with urban youth.

5.3 Table 1: Summary statistics

Read. Rural women in the main cohorts have less schooling than rural men. Female middle school completion is 41.0% versus 62.6% for males; high school completion is 9.3% versus 16.7%. Women also have lower formal labor participation and weaker financial independence.

Economic message. The baseline gender gaps are large, so changes in education, labor participation, and own-work income are meaningful measures of empowerment.

5.4 Figure 3: Raw relationship between SDY exposure and female education



(A) **Middle school (Treat)**

(B) **Middle school (Control)**

(C) **High school (Treat)**

(D) **High school (Control)**

Figure 3: Raw relationship between SDY share and female educational attainment.

Read. For treatment cohorts, counties with higher SDY shares have higher female middle school and high school completion. For control cohorts, the relationship is weaker.

Economic message. The raw data already show the DID logic: SDY intensity matters more for cohorts who were young enough to be exposed in school.

5.5 Table 2: Main education, labor, and financial independence results

Table 2
Effect of SDY on Education, Employment, and Financial independence.

Dependent variable	(1)	(2)	(3)	(4)	(5)	(6)
	Education attainment				Labor participation	
	Middle school completion		High school completion		Working in 2000	
	Female	Male	Female	Male	Female	Male
$SDY \times Treat$	0.434***	0.072	0.647***	0.490***	0.694***	0.220***
	(0.131)	(0.089)	(0.063)	(0.067)	(0.072)	(0.033)
Female - Male[a]	0.362***		0.157***		0.474***	
p-value	[0.000]		[0.001]		[0.000]	
Observations	1,264,842	1,284,398	1,264,842	1,284,398	1,264,842	1,284,398
R-squared	0.209	0.164	0.064	0.060	0.119	0.027
Control-group mean	0.215	0.453	0.046	0.114	0.812	0.953
Dependent variable	(7)	(8)	(9)	(10)	(11)	(12)
	Labor participation		Main source of income			
	Weekly working hours		Own work		Family support	
	Female	Male	Female	Male	Female	Male
$SDY \times Treat$	24.571***	7.019	0.651***	0.210**	−0.434***	−0.104
	(5.945)	(4.625)	(0.120)	(0.082)	(0.113)	(0.072)
Female - Male[a]	17.552***		0.441***		−0.330***	
p-value	[0.003]		[0.000]		[0.003]	
Observations	312,350	304,241	312,350	304,241	312,350	304,241
R-squared	0.220	0.157	0.171	0.087	0.158	0.075
Control-group mean	25.496	36.278	0.676	0.871	0.290	0.086
Controls	✓	✓	✓	✓	✓	✓
County FE	✓	✓	✓	✓	✓	✓
Province-cohort FE	✓	✓	✓	✓	✓	✓

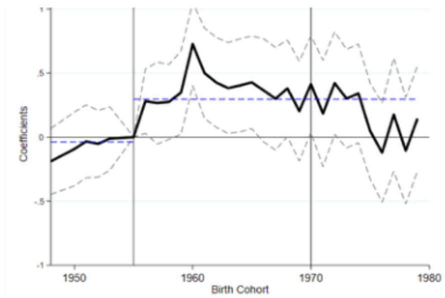
Notes: This table reports estimated SDY's effect on educational attainments, labor market outcomes, and financial independence. The unit of observation is at the individual level. Data are sourced from Population Census 2000 in columns 1–6 and Population Census 2010 in columns 7–12. The sample includes all individuals born between 1946 and 1969. The regression model is specified in Eq. (1). SDY is the county-level share of received SDYs during 1968–1977 in the pre-movement local population in 1964. $Treat$ equals one for individuals born between 1956–1969, and zero if born between 1946–1955. Control variables include indicators of ethnic groups. Robust standard errors clustered at the county level are presented in parentheses. ***p< 0.01, **p <0.05, *p < 0.1.

Figure 4: Baseline effects on education, employment, and financial independence.

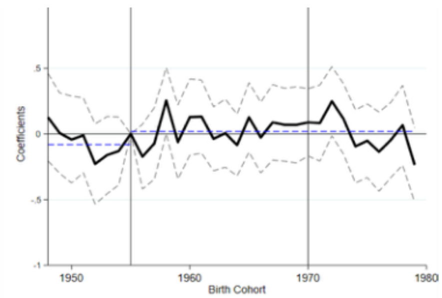
Read. SDY exposure increases rural female middle school completion, high school completion, labor participation, weekly working hours, and reliance on own work as income. Female effects are generally larger than male effects. For example, a two-pp increase in SDY share raises female middle school completion by about 0.868 pp and high school completion by about 1.29 pp.

Economic message. SDYs improved human capital and labor-market outcomes, but the stronger female effects imply that the movement also narrowed gender gaps.

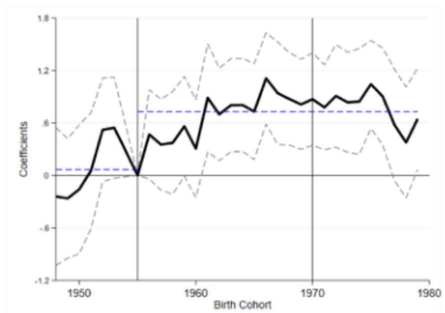
5.6 Figure 4: Event study for individual outcomes



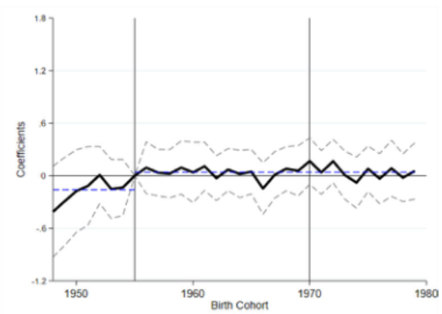
(A) **Middle school completion (female)**



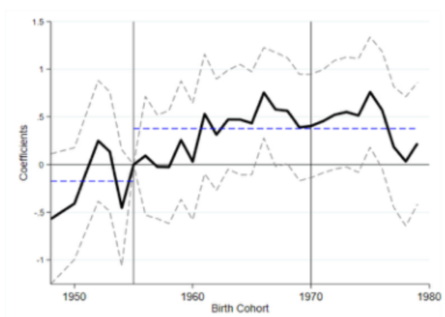
(B) **Middle school completion (male)**



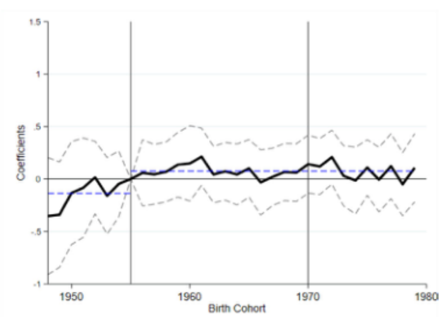
(C) **Labor participation (female)**



(D) **Labor participation (male)**



(E) **Financial Independence (female)**



(F) **Financial Independence (male)**

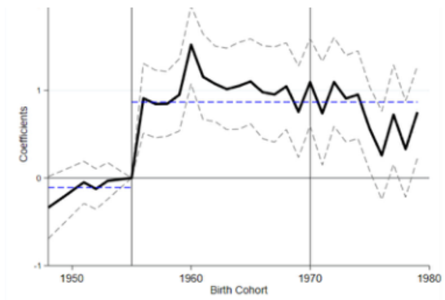
analysis on SDY's effect on rural individual's education, labor market participation, and financial independence.
ots the event study coefficients and corresponding 95% confidence intervals for SDY's impact on middle school completion rates (Panels A and B), labor

Figure 5: Event-study effects on individual education, labor participation, and financial independence.

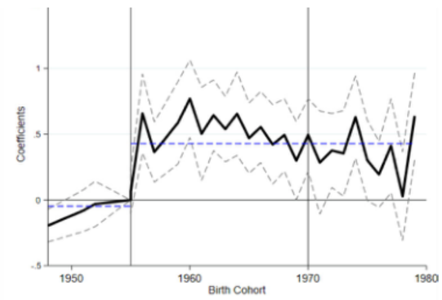
Read. Pre-treatment cohort coefficients are small. Effects become more positive for treatment cohorts, especially for rural females.

Economic message. The timing supports the identification strategy. Outcomes change for exposed cohorts rather than trending differently before the movement.

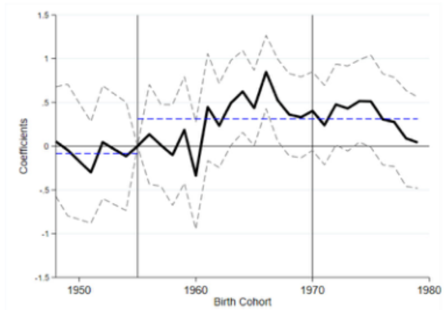
5.7 Figure 5: Event study for gender gaps



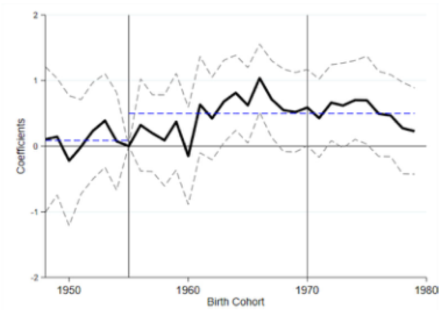
(A) **Middle school completion (ratio)**



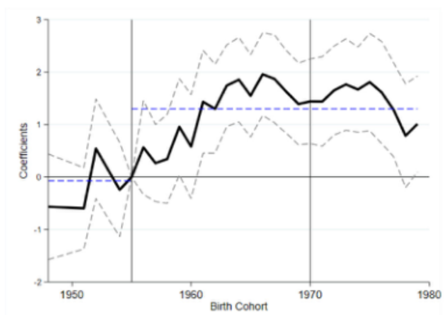
(B) **Middle school completion (diff)**



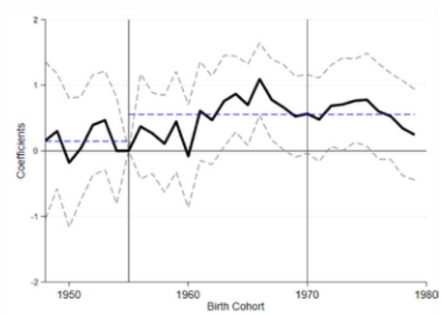
(C) **Labor participation (ratio)**



(D) **Labor participation (diff)**



(E) **Financial independence (ratio)**



(F) **Financial independence (diff)**

analysis on SDY's effect on rural gender gap in education, labor market participation, and financial independence.
lots the event study coefficients to analyze SDY's impact on gender gaps in education, labor market participation, and financial independence. The

Figure 6: Event-study effects on rural gender gaps.

Read. Gender-gap measures show no strong pre-trend for older cohorts. Gender equality improves for treatment cohorts in high-SDY counties.

Economic message. This figure directly confirms the gender-equality interpretation: SDYs did not merely raise outcomes for everyone; they disproportionately improved women's relative position.

5.8 Table 3: Marriage and fertility

Table 3
Effect of SDY on female marriage and fertility.

Dependent variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Marriage						Fertility
	First marry age if ever married		Ever divorced		Never married		# of children
	Female	Male	Female	Male	Female	Male	Female
Panel A: Baseline setting							
$SDY \times Treat$	1.587**	0.096	0.070***	0.049**	0.018***	0.047**	−1.801***
	(0.691)	(0.526)	(0.020)	(0.023)	(0.006)	(0.020)	(0.303)
Female - Male[a]	1.491***		0.021		−0.029		
p-value	[0.007]		[0.453]		[0.165]		
Panel B: Adding education as control							
$SDY \times Treat$	1.247**	0.060	0.071***	0.050**	0.018***	0.064***	−1.799***
	(0.626)	(0.520)	(0.020)	(0.023)	(0.006)	(0.021)	(0.302)
Schooling years	0.179***	0.030***	−0.001***	−0.001***	0.000	−0.016***	−0.014***
	(0.002)	(0.003)	(0.000)	(0.000)	(0.000)	(0.000)	(0.001)
Female - Male[b]	1.187**		0.021		−0.046**		
p-value	[0.023]		[0.454]		[0.030]		
Observations	1,261,433	1,222,035	1,264,842	1,284,398	1,264,842	1,284,398	311,688
Control-group mean	21.616	24.185	0.004	0.013	0.001	0.046	2.658
Control	✓	✓	✓	✓	✓	✓	✓
County FE	✓	✓	✓	✓	✓	✓	✓
Province-cohort FE	✓	✓	✓	✓	✓	✓	✓

Figure 7: Effects on marriage and fertility decisions.

Read. SDY exposure delays women's first marriage, increases the probability of ever divorcing and never marrying, and reduces the number of children. These effects remain after controlling for years of schooling.

Economic message. Female empowerment extends beyond education. Rural women exposed to SDYs make different family-formation decisions, suggesting greater autonomy in marriage and fertility.

5.9 Table 4: Political empowerment

Table 4
Effect of SDY on female political empowerment.

Dependent variable	(1)	(2)	(3)	(4)	(5)	(6)
	Party member	Political participation via			Believing Gov's role as	
		Civic activities	Petitioning	Strikes and boycotts	Social security	Law and order
Panel A: Baseline setting						
$SDY \times Treat \times Female$	2.534***	10.303***	2.186**	−4.218***	6.402***	6.520*
	(0.745)	(2.153)	(0.868)	(1.367)	(2.402)	(3.552)
Panel B: Adding education as control						
$SDY \times Treat \times Female$	1.175*	10.154***	2.170**	−4.237***	6.509***	6.629*
	(0.638)	(2.165)	(0.857)	(1.382)	(2.426)	(3.566)
Schooling years	0.016***	−0.001	0.000	−0.008	0.017	−0.008
	(0.002)	(0.011)	(0.005)	(0.007)	(0.013)	(0.009)
$Schooling\ years \times Female$	0.010***	0.017	0.001	0.014	0.023	0.010
	(0.002)	(0.013)	(0.007)	(0.009)	(0.015)	(0.010)
Observations	4905	3351	2242	1357	3066	3238
Control-group mean	0.074	2.450	0.202	0.182	3.291	3.178
Gender-specific controls	✓	✓	✓	✓	✓	✓
Gender-County-Cohort FEs	✓	✓	✓	✓	✓	✓
Gender-Survey-Wave FEs		✓	✓	✓	✓	✓

Figure 8: Effects on female political empowerment.

Read. Relative to men, exposed rural women are more likely to be Party members and show stronger support for some forms of political and civic participation. The effects survive schooling controls.

Economic message. SDY exposure affected social and political agency, not only market outcomes. This is important because political participation is a distinct dimension of empowerment.

5.10 Table 5: Preferences, self-perceptions, and social values

Table 5
Effect of SDY on Female preference, Perception, and Social values.

Dependent variable	(1) Value formal employment	(2) Value financial independence	(3) Overcoming challenges indep.	(4) Self-confidence	(5) Willingness to take risk
Panel A: Baseline setting					
$SDY \times Treat \times Female$	2.889***	2.977***	2.199***	1.205***	1.409**
	(0.468)	(0.566)	(0.575)	(0.328)	(0.696)
Panel B: Adding education as control					
$SDY \times Treat \times Female$	1.662***	1.653***	1.929***	0.929***	0.912
	(0.479)	(0.556)	(0.637)	(0.348)	(0.707)
Schooling years	0.002***	0.002***	0.000	0.000	0.001**
	(0.000)	(0.000)	(0.000)	(0.000)	(0.001)
Schooling years $\times$ *Female*	0.003***	0.003***	0.001	0.001	0.001
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Observations	7389	7386	7391	7387	7363
R-squared	0.278	0.293	0.271	0.284	0.360
Control-group mean	0.773	0.813	0.843	0.834	0.215
Gender-specific controls	✓	✓	✓	✓	✓
Gender-County-Cohort FE	✓	✓	✓	✓	✓

Notes: This table reports results testing for the impact of SDYs on female political participation, self-perception, and social values. Data are sourced from CHIP 2008. The sample includes all cohorts born between 1946 and 1969. Dependent variables are self-rated importance of formal employment (Column 1), financial independence (Column 2), overcoming challenges independently (Column 3), self-confidence (Column 4), and willingness to take risks (Column 5). Responses for Columns 1 to 5 are initially ranked on different scales and have been normalized to a range of 0 to 1, where a higher value signifies greater agreement with the statement. SDY is the county-level SDY population share, defined as the number of received SDYs during 1968–1977 divided by the pre-movement local population in 1964. $Treat$ is a dummy variable indicating whether the individual was born between 1956 and 1969. Control variables include indicators of ethnic groups. Robust standard errors clustered at the county level are presented in parentheses. ***p< 0.01, **p <0.05, *p < 0.1.

Figure 9: Effects on preferences, self-perceptions, and social values.

Read. Exposed rural women place more value on formal employment, financial independence, overcoming challenges independently, self-confidence, and willingness to take risk. These effects are not eliminated by schooling controls.

Economic message. The table is central for the ideology channel. The movement changed women's stated aspirations and self-perceptions, consistent with transmission of gender-equal values.

5.11 Mechanism and welfare evidence from appendix analyses

Read. Mechanism tests show that longer school exposure produces larger education and labor-market gains, while gender-equal attitudes respond even with shorter exposure. Out-of-school analyses show that young rural women who were close in age to SDYs also changed labor, marriage, fertility, and attitude outcomes. Effects are stronger in tea-growing areas, where female SDYs and rural women likely interacted more, but not in orchard areas, a placebo setting with more male labor.

Economic message. The paper's mechanism is not only classroom human capital. Social interaction and role-model exposure appear to transmit norms and aspirations. Welfare analyses also show later-life improvements in mental health, happiness, and life satisfaction.

6 Short summary

- **Question:** Can internal population mobility transmit gender-equal values and empower women in traditional rural societies?
- **Setting:** China's Send-down Movement, which relocated over 16 million urban youth to rural counties during the late 1960s and 1970s.
- **Design:** Cohort DID using county SDY population share interacted with whether rural individuals were primary-school age during the movement.
- **Main result:** Greater SDY exposure improved rural women's education, formal employment, financial independence, marriage autonomy, fertility decisions, political participation, and gender-equal beliefs.
- **Gender result:** Effects were generally larger for women than for men, so SDYs narrowed gender gaps in education, labor-market outcomes, and financial autonomy.
- **Mechanism:** Human capital matters, but the evidence also points to transmission of gender-equal ideologies through school and out-of-school social interactions.
- **Credibility:** Event studies show no strong pre-trends, and the results survive extensive checks for pre-movement characteristics, historical shocks, later migration, urban spillovers, and alternative cohort definitions.

7 Conclusion

7.1 Core conclusion

The paper shows that the Send-down Movement had long-lasting effects on rural female empowerment in China. Rural girls exposed to more urban youth during primary-school age attained more education, worked more in formal labor markets, became more financially independent, delayed marriage, had fewer children, participated more in politics, and adopted more gender-equal beliefs.

7.2 Economic intuition

Population mobility can transmit ideas as well as people. Urban SDYs entered rural communities with higher education and more progressive gender ideologies. Their roles as teachers, coworkers, and peers exposed rural females to new information, new role models, and new beliefs about what women could do in school, work, family, and public life.

7.3 Takeaway

The paper's broad lesson is that female empowerment can be accelerated through social contact between groups with different norms. Even temporary population integration can have persistent effects when it reaches children and young adults at formative ages.